HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and maintained **production backend services** supporting **AI agents** and **developer assistants**, ensuring reliability and performance in a high-traffic environment.
- Owned **service architecture** and infrastructure decisions, leading to improved system reliability and maintainability for critical AI tooling.
- Defined and monitored **service reliability objectives (SLOs)**, tracking error budgets and prioritizing improvements in collaboration with SRE teams.
- Participated in the team's **on-call rotation**, investigating production incidents, coordinating resolutions, and developing **preventive solutions** and runbooks.
- Managed operational support requests, identifying recurring issues and implementing long-term engineering improvements to enhance system stability.
- Designed and maintained **authentication and authorization systems** for AI agents, including **scoped tokens** and secure access management.
- Established comprehensive **observability** through metrics, distributed tracing, logging, dashboards, and alerting systems for AI tooling infrastructure.

SKILLS

Backend Development – TypeScript, Go, Node.js, NestJS, Java, Spring Boot, REST APIs

AI & ML Infrastructure – AI agents, AI assistants, Model Context Protocol (MCP), ML Ops

Databases – PostgreSQL, Relational Databases

Cloud & DevOps – AWS, Docker, Kubernetes, CI/CD, Infrastructure as Code (IaC), Jenkins

Observability & Reliability – SLOs, Error Budgets, Metrics, Tracing, Logging, Alerting, PagerDuty, Grafana, OpenTelemetry

Security & Authentication – OAuth, OpenID Connect (OIDC), JWT, API Authorization, Permissions, Token Scopes

PROJECTS

AI Agent Orchestration Service

- Developed a **backend service** in **TypeScript** (**NestJS**) to orchestrate AI agent tasks, managing state and dependencies for complex workflows.
- Implemented **rate limiting** and **capacity controls** using **PostgreSQL** to manage AI agent traffic and safeguard against unexpected spikes.
- Integrated with internal **authentication systems** to secure agent access, handling token validation and permission checks for machine clients.

Production Observability Platform Enhancement

- Engineered enhancements to the **production observability platform**, integrating **distributed tracing** and refining **alerting** mechanisms for critical AI services.
- Developed automated scripts using **Python** to parse logs and generate performance metrics, improving incident detection time by **~18%**.
- Containerized new monitoring agents with **Docker** and deployed them to a **Kubernetes** cluster, ensuring seamless integration with the existing infrastructure.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)